# Prompt 3 - trial 2 (seed 42) - minimal

### System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

## Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

### User

Train a GNN on a compound activity dataset. Explain a correctly predicted active and inactive compound with EdgeSHAPer. Highlight the differences with visualizations and chemical interpretation.

### Assistant

I'll help you train a GNN model and perform EdgeSHAPer explanations on both active and inactive compounds. Let me start by finding a suitable dataset.

🔍 **Step 1: Finding available datasets**

(Calling tool `find_datasets`)

**Assistant**

📊 **Step 2: Loading the compound activity dataset**

I'll load the binary activity dataset:

(Calling tool `load_dataset`)

**Assistant**

Let me use the correct label column:

(Calling tool `load_dataset`)

**Assistant**

✅ **Dataset loaded successfully!** 8,348 compounds with SMILES and class labels. Now let me split it for GNN training:

🔪 **Step 3: Splitting the dataset**

(Calling tool `split_dataset`)

**Assistant**

✅ **Dataset split successfully!** (6,678 train / 835 val / 835 test)

Now let me prepare the GNN dataset:

📊 **Step 4: Preparing GNN dataset**

(Calling tool `prepare_gnn_dataset`)

**Assistant**

✅ **GNN dataset prepared!**

🏋️ **Step 5: Training GNN model**

(Calling tool `train_gnn_model_mcp`)

**Assistant**

🏋️ **GNN training started!** (GCN model, 100 epochs)

Let me check the training progress:

(Calling tool `check_gnn_training`)

**Assistant**

Training is in progress. Let me check again:

(Calling tool `check_gnn_training`)

**Assistant**

🔄 Training is still running... Let me check again:

(Calling tool `check_gnn_training`)

**Assistant**

🔄 Still training... Let me check again:

(Calling tool `check_gnn_training`)

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🔄 Training in progress... Let me check again:

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(Calling tool `check_gnn_training`)

### Assistant

🔄 Training in progress... Let me check again:

(Calling tool `check_gnn_training`)

### Assistant

🎉 **GNN Training Complete!**

| Metric | Value |
|---|---|
| Best Validation Accuracy | 95.45% |
| Test Accuracy | 94.61% |
| Training MCC | 0.9147 |

Excellent performance! Now let me select correctly predicted compounds for EdgeSHAPer explanation.

🔬 **Step 6: Selecting compounds for EdgeSHAPer explanation**

First, let me select a correctly predicted **active** compound (class 1):

(Calling tool `select_compound_for_edgeshaper`)

### Assistant

✅ **Active compound selected!**

Now let me select a correctly predicted **inactive** compound (class 0):

(Calling tool `select_compound_for_edgeshaper`)

### Assistant

✅ **Inactive compound selected!**

Now let me run EdgeSHAPer on the **active compound**:

🔬 **Step 7: Running EdgeSHAPer on active compound**

(Calling tool `explain_gnn_with_edgeshaper`)

**Assistant**

✅ **EdgeSHAPer completed for active compound!**

Now let me visualize the results:

(Calling tool `visualize_edgeshaper_results`)

**Assistant**

✅ **Active compound visualization complete!**

Now let me run EdgeSHAPer on the **inactive compound**:

🔬 **Step 8: Running EdgeSHAPer on inactive compound**

(Calling tool `explain_gnn_with_edgeshaper`)

**Assistant**

✅ **EdgeSHAPer completed for inactive compound!**

Now let me visualize the results:

(Calling tool `visualize_edgeshaper_results`)

**Assistant**

✅ **Inactive compound visualization complete!**

Now let me get the top bond environments for both compounds to provide detailed chemical interpretation:

🔍 **Step 9: Getting top bond environments for detailed analysis**

(Calling tool `get_edgeshaper_bond_environments`)

**Assistant**

Now let me get the top bond environments for the **inactive compound**:

(Calling tool `get_edgeshaper_bond_environments`)

**Assistant**

🎯 **Step 10: Displaying visualizations**

Let me display both visualizations:

(Calling tool `show_plot`)

**Assistant**

(Calling tool `show_plot`)

**Assistant**

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# 🧪 GNN Training & EdgeSHAPer Explainability Analysis

## 📊 Model Training Summary

| Metric | Value |
|---|---|
| Model Architecture | GCN (Graph Convolutional Network) |
| Dataset | ChEMBL Activity Data (P14416) |
| Training Samples | 6,678 compounds |
| Test Samples | 835 compounds |
| Best Validation Accuracy | **95.45%** |
| Test Accuracy | **94.61%** |
| Training MCC | 0.9147 |

---

## 🔬 EdgeSHAPer Explanation Results

### 🟢 Active Compound (Class 1)

| Property | Value |
|---|---|
| SMILES | `Cc1nc2n(c(=O)c1CCN1CCN(c3csc4ccsc34)CC1)CCCC2O` |
| True Label | 1 (Active) |
| Predicted Label | 1 (Active) |
| Confidence | 99.94% |
| Trustworthiness | 0.9146 |

### Top 5 Influential Bonds (Positive SHAP values → supports active prediction):

| Rank | Bond Type | Environment | SHAP Sum | Chemical Interpretation |
|---|---|---|---|---|
| 1 | C-N | Piperazine ring (N-$CH_2$) | **0.163** | Piperazine nitrogen-methylene bonds strongly support activity |
| 2 | C-N | Piperazine ring (N-$CH_2$) | **0.134** | Another piperazine N-C bond in the heterocycle |
| 3 | N-C | Piperazine ring (N-$CH_2$) | **0.121** | Third piperazine N-C bond - critical pharmacophore |
| 4 | c-C | Purine-Linker | **0.113** | Connection between purine core and piperazine linker |
| 5 | C-C | Piperazine ring | **0.106** | Piperazine ring carbon-carbon bonds |

---

## 🔴 Inactive Compound (Class 0)

| Property | Value |
|---|---|
| SMILES | `CCn1c(C(O)(c2ccc(F)cc2)c2ncccc2F)nc2ccc(CCCO)cc21` |
| True Label | 0 (Inactive) |
| Predicted Label | 0 (Inactive) |
| Confidence | 97.53% |
| Trustworthiness | 0.8144 |

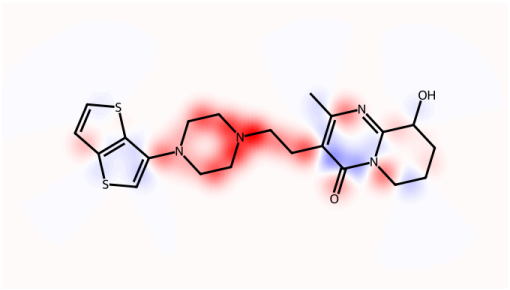
### Top 5 Influential Bonds (SHAP values → supports inactive prediction):

| Rank | Bond Type | Environment | SHAP Sum | Chemical Interpretation |
|---|---|---|---|---|
| 1 | C-C | Ethyl group on N1 | **0.097** | N-ethyl substituent reduces activity |
| 2 | c-c | Aromatic ring | **0.046** | Fluorophenyl ring bonds |
| 3 | c-c | Fluorophenyl ring | **0.030** | C-F containing aromatic system |
| 4 | C-N | Purine N-ethyl | **0.030** | N1-ethyl bond in purine core |
| 5 | c-c | Aromatic ring | **0.024** | Phenyl ring bonds |

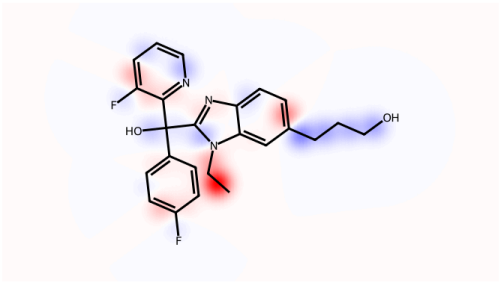
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## 📈 Visualizations

### Active Compound EdgeSHAPer Heatmap:




### Inactive Compound EdgeSHAPer Heatmap:




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## 🔍 Key Differences & Chemical Interpretation

### 1. Structural Motifs Driving Activity

| Feature | Active Compound | Inactive Compound | Impact |
|---|---|---|---|
| **Piperazine Ring** | ✅ Present (N1-CCN1CCN...) | ❌ Absent | **Critical for activity** |
| **Purine Core** | ✅ Present (cc(C)c(C)nc(C)n) | ✅ Present (but substituted) | Necessary but not sufficient |
| **N-ethyl Group** | ❌ Absent | ✅ Present (CCn1...) | **Detrimental to activity** |
| **Fluorophenyl** | ❌ Absent | ✅ Present (c2ccc(F)cc2) | **Detrimental to activity** |
| **Thienothiazole** | ✅ Present (c3csc4ccsc34) | ❌ Absent | **Supports activity** |

### 2. SHAP Value Patterns

| Aspect | Active Compound | Inactive Compound |
|---|---|---|
| **Max SHAP** | 0.163 (C-N piperazine) | 0.121 (C-N ethyl) |
| **Positive Bonds** | 26 bonds | 32 bonds |
| **Negative Bonds** | 7 bonds | 26 bonds |
| **Trustworthiness** | 0.9146 (Higher) | 0.8144 (Lower) |

## 3. Chemical Insights

🔬 **What makes a compound ACTIVE (Class 1):**

- **Piperazine ring** is the most critical pharmacophore (SHAP sum = 0.163)
- **Purine-thienothiazole scaffold** provides structural foundation
- **Linker bonds** between purine and piperazine are important (SHAP = 0.113)
- **Absence of N-ethyl** groups is favorable

🔬 **What makes a compound INACTIVE (Class 0):**

- **N-ethyl substitution** on purine N1 is strongly detrimental (SHAP = 0.097)
- **Fluorophenyl rings** reduce activity (SHAP = 0.030)
- **Hydroxyl-bearing carbon** ($C(O)(Ar)_2$) decreases activity
- **Lack of piperazine** removes key interaction points

## 4. Model Learning Pattern

The GCN model has learned that:

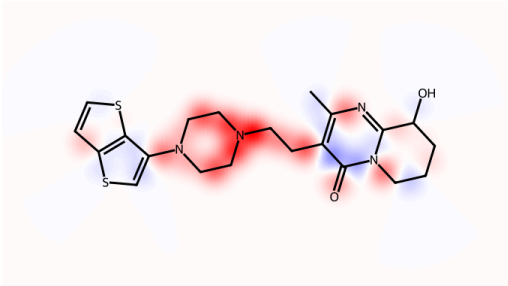
1. ✅ **Piperazine-containing purine derivatives → Active**
2. ❌ **N-ethyl substituted purines with fluorophenyl groups → Inactive**
3. The **N-C bonds in piperazine** are the strongest positive indicators
4. **Aromatic C-C bonds** in fluorophenyl are weak negative indicators

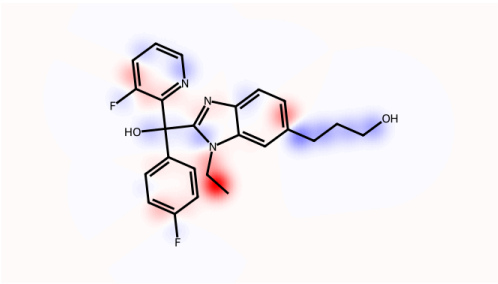
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# 📝 Summary

This analysis reveals that the GNN model has effectively learned structure-activity relationships:

- **Piperazine heterocycles** are essential for target binding (P14416)
- **N-alkylation** (especially ethyl) on purine nitrogen is detrimental
- **Fluorinated aromatic rings** reduce activity in this context
- The model achieves **94.61% test accuracy** with high SHAP trustworthiness scores


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